UNION PUBLIC SERVICE COMMISSION (UPSC)

CDS GENERAL KNOWLEDGE — SUBJECT 4: GENERAL SCIENCE (PHYSICS, CHEMISTRY, BIOLOGY)

OFFICIAL MASTER ANSWER KEY & COMPREHENSIVE EXPLANATIONS

Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
1	(A)	7	(B)	13	(B)	19	(A)	25	(A)	31	(A)	37	(B)	43	(A)	49	(A)	55	(A)
2	(B)	8	(A)	14	(A)	20	(B)	26	(C)	32	(A)	38	(B)	44	(C)	50	(A)	56	(C)
3	(A)	9	(B)	15	(A)	21	(B)	27	(C)	33	(A)	39	(B)	45	(A)	51	(C)		
4	(C)	10	(A)	16	(B)	22	(C)	28	(A)	34	(B)	40	(B)	46	(A)	52	(A)		
5	(A)	11	(C)	17	(A)	23	(C)	29	(A)	35	(A)	41	(C)	47	(C)	53	(A)		
6	(A)	12	(A)	18	(C)	24	(A)	30	(C)	36	(A)	42	(A)	48	(A)	54	(C)		

COMPREHENSIVE CONCEPTUAL SOLUTIONS & RATIONALES:

Q.1 — Official Correct Answer: (A) Malaria

Question: A female Anopheles mosquito acts as the biological vector for which one of the following diseases?
Solution Rationale: Female Anopheles mosquitoes transmit Plasmodium parasites causing Malaria. Dengue, Chikungunya, and Yellow fever are transmitted by Aedes mosquitoes.

Q.2 — Official Correct Answer: (B) Concave lenses

Question: A person suffering from myopia (near-sightedness) is advised to wear lenses of:
Solution Rationale: Myopia occurs when the eyeball is too elongated or cornea too curved, causing distant rays to focus in front of the retina. A diverging (concave) lens spreads the incoming rays so the image forms precisely on the retina.

Q.3 — Official Correct Answer: (A) 6 kg

Question: An object has a mass of 6 kg on the surface of the Earth. When taken to the Moon (where acceleration due to gravity is 1/6th that of Earth), its mass will be:
Solution Rationale: Mass is an intrinsic property representing the amount of matter in a body and remains constant everywhere (6 kg). Only the weight (W = mg) decreases to 1/6th on the Moon.

Q.4 — Official Correct Answer: (C) Both 1 and 2

Question: Consider the following statements regarding the Chief of Defence Staff (CDS) of the Indian Armed Forces: 1. The CDS functions as the permanent Chairman of the Chiefs of Staff Committee. 2. The CDS acts as the principal military adviser to the Defence Minister on tri-service matters. Which of the statements given above is/are correct?
Solution Rationale: Both statements are correct: The CDS is a 4-star General who heads the Department of Military Affairs (DMA), chairs the Chiefs of Staff Committee, and acts as principal military adviser to the Raksha Mantri.

Q.5 — Official Correct Answer: (A) Trisomy of Chromosome 21

Question: Down's syndrome, a genetic condition causing developmental and intellectual delays in humans, is caused by which chromosomal abnormality?
Solution Rationale: Down's syndrome is caused by non-disjunction of chromosome 21 during meiosis, resulting in cells having 47 chromosomes instead of 46 (Trisomy 21).

Q.6 — Official Correct Answer: (A) Gas

Question: For the same rise in temperature, which of the three states of matter undergoes the greatest volumetric thermal expansion?
Solution Rationale: Gases exhibit the highest coefficient of thermal expansion because intermolecular forces are minimal, allowing particles to move far apart when heated.

Q.7 — Official Correct Answer: (B) Exosmosis leading to plasmolysis

Question: If excess chemical fertilizer is applied to a potted plant without adding adequate water, the plant wilts and dies because of:
Solution Rationale: High concentration of fertilizer salts creates a hypertonic external solution around the roots. Water is drawn out from plant cells via exosmosis, causing cellular plasmolysis and wilting.

Q.8 — Official Correct Answer: (A) Ratio of dependent population (0-14 and 65+ years) to working-age population (15-64 years)

Question: In demographic studies and development economics, the Dependency Ratio of a nation is mathematically defined as:
Solution Rationale: The dependency ratio measures the age-population ratio of typically dependent citizens (under 15 and over 64) relative to the productive working-age population.

Q.9 — Official Correct Answer: (B) Pyloric sphincter

Question: In the human digestive system, the opening of the stomach into the small intestine (duodenum) is regulated by the:
Solution Rationale: The pyloric sphincter is a band of smooth muscle at the junction between the pylorus of the stomach and the duodenum of the small intestine that controls the flow of chyme.

Q.10 — Official Correct Answer: (A) He

Question: In the light of this statement
Solution Rationale: Official UPSC CDS Rationale: Tested in CDS General Knowledge (General Science (Physics, Chemistry, Biology)).

Q.11 — Official Correct Answer: (C) Golgi bodies

Question: Lysosome is formed from which one of the following cell organelles?
Solution Rationale: Lysosomes are membrane-bound organelles containing digestive enzymes that are packaged and budded off from the Golgi apparatus/bodies.

Q.12 — Official Correct Answer: (A) 15 judges

Question: The International Court of Justice (ICJ), the principal judicial organ of the United Nations based at The Hague, is composed of how many judges elected for a nine-year term?
Solution Rationale: The ICJ consists of 15 judges elected by the UN General Assembly and Security Council for 9-year terms. Five judges are elected every three years.

Q.13 — Official Correct Answer: (B) Vishakha Mulye

Question: The Reserve Bank of India has recently constituted a high-level task force on Public Credit Registry (PCR) to suggest a road map for developing a transparent, comprehensive and near-real-time PCR for India. The task force is headed by
Solution Rationale: Official UPSC CDS Solution: Assessed under General Science (Physics, Chemistry, Biology) according to UPSC CDS standards.

Q.14 — Official Correct Answer: (A) Archimedes' Principle of Buoyancy

Question: The fundamental physical law which explains why a heavy steel ship floats on the ocean while a small iron needle sinks is:
Solution Rationale: Archimedes' Principle states that a body immersed in a fluid is buoyed up by a force equal to the weight of fluid displaced. A hollow ship displaces a volume of water whose weight exceeds the ship's total weight.

Q.15 — Official Correct Answer: (A) Inner ear (vestibular apparatus)

Question: The human body's primary sensory organ responsible for maintaining balance, posture, and equilibrium is located in the:
Solution Rationale: The vestibular apparatus in the inner ear, consisting of the three semicircular canals, utricle, and saccule, detects head motion and gravity to maintain balance and equilibrium.

Q.16 — Official Correct Answer: (B) 2 and 3 only

Question: The kinetic energy of an object is always conserved.
Solution Rationale: Official UPSC CDS Solution: Assessed under General Science (Physics, Chemistry, Biology) according to UPSC CDS standards.

Q.17 — Official Correct Answer: (A) Keratin proteins and amino acids

Question: The natural plant dye Henna (Lawsone) imparts an orange-red stain to human skin and hair by reacting primarily with:
Solution Rationale: The active colorant in henna is lawsone (hennotannic acid), which binds strongly through Michael addition to keratin proteins found in hair, nails, and the epidermis of the skin.

Q.18 — Official Correct Answer: (C) 1 and 2 only

Question: Vitamin A ß-carotene ß-carotene Vitamin A engineered 'Golden in
Solution Rationale: Official UPSC CDS Rationale: Tested in CDS General Knowledge (General Science (Physics, Chemistry, Biology)).

Q.19 — Official Correct Answer: (A) Red and Blue

Question: Wavelengths of which of the following colors of the visible light spectrum are maximally absorbed by chlorophyll in green plants for photosynthesis?
Solution Rationale: Chlorophyll a and b show maximum absorption in the blue (430-450 nm) and red (640-660 nm) regions of the visible spectrum, while reflecting green light (giving plants their green appearance).

Q.20 — Official Correct Answer: (B) Moderator and coolant

Question: What is 'heavy water' (D2O) used as in nuclear power reactors?
Solution Rationale: Heavy water (deuterium oxide, D2O) contains deuterium atoms and is used primarily as a neutron moderator (slowing down fast neutrons) and as a coolant in Pressurized Heavy Water Reactors (PHWRs).

Q.21 — Official Correct Answer: (B) A repository for storage and management of data where mechanical, electrical, and computer systems are optimized for maximum energy efficiency and minimal environmental footprint

Question: What is a 'Green Data Centre'?
Solution Rationale: A green data centre is an enterprise storage repository where lighting, electrical, cooling, and compute systems are engineered to achieve maximum power usage effectiveness (PUE) and minimal carbon emissions.

Q.22 — Official Correct Answer: (C) Nagas

Question: What is that particular tribal group?
Solution Rationale: Official UPSC CDS Rationale: Tested in CDS General Knowledge (Defence, Military & Current Affairs).

Q.23 — Official Correct Answer: (C) Preempt social agitations by enacting sensitive laws

Question: What is the minimum mandate of a democratic government ?
Solution Rationale: Official UPSC CDS Rationale: Tested in CDS General Knowledge (Defence, Military & Current Affairs).

Q.24 — Official Correct Answer: (A) Calcium carbonate (Aragonite)

Question: What is the primary chemical compound that constitutes natural pearls and sea shells?
Solution Rationale: Pearls and mollusk shells consist primarily of calcium carbonate (CaCO3, mostly in the form of aragonite crystals) deposited in concentric layers held together by the organic protein conchiolin.

Q.25 — Official Correct Answer: (A) Fats/Vegetable oils and Caustic soda (Sodium hydroxide)

Question: Which among the following are the two primary raw materials used in the saponification process to manufacture soap?
Solution Rationale: Soap is produced by saponification, the alkaline hydrolysis of triglycerides (vegetable oils or animal fats) with a strong base such as sodium hydroxide (caustic soda) or potassium hydroxide.

Q.26 — Official Correct Answer: (C) Soya bean oil

Question: Which among the following oils has the maximum protein •content ?
Solution Rationale: Official UPSC CDS Rationale: Tested in CDS General Knowledge (General Science (Physics, Chemistry, Biology)).

Q.27 — Official Correct Answer: (C) Mitochondria

Question: Which cell organelle is famously known as the 'Powerhouse of the Cell' due to ATP production?
Solution Rationale: Mitochondria produce adenosine triphosphate (ATP) through cellular respiration and oxidative phosphorylation, thereby serving as the powerhouse of eukaryotic cells. Lysosomes are suicide bags.

Q.28 — Official Correct Answer: (A) 1 only

Question: Which of the following statements relating to the Non-Aligned Movement (NAM) is/are correct? 1. It was founded to maintain independence from rival Cold War power blocs. 2. Military alliances formed a core component of NAM. Select the correct answer using the code given below:
Solution Rationale: Statement 1 is correct: NAM originated at the 1955 Bandung Conference and 1961 Belgrade Summit. Statement 2 is incorrect: NAM was explicitly non-military and rejected joining Cold War military alliances like NATO or the Warsaw Pact.

Q.29 — Official Correct Answer: (A) Argon

Question: Which of the following noble gases is present in the highest abundance by volume in dry atmospheric air?
Solution Rationale: Argon is the third most abundant gas in dry air (approx. 0.934% by volume). Other noble gases like Neon (0.0018%), Helium (0.0005%), and Krypton (0.0001%) are trace elements.

Q.30 — Official Correct Answer: (C) It can be applied automatically by local district magistrates without any disturbed area notification

Question: Which of the following statements regarding the Armed Forces (Special Powers) Act (AFSPA) is not correct?
Solution Rationale: AFSPA cannot be invoked arbitrarily by district magistrates; an area must be formally declared a 'disturbed area' by the Central Government, Governor, or UT Administrator under Section 3 of the Act.

Q.31 — Official Correct Answer: (A) Kala-azar (Visceral leishmaniasis)

Question: Which of the following infectious diseases is caused by a protozoan parasite and transmitted by the bite of the sandfly?
Solution Rationale: Kala-azar (Visceral leishmaniasis) is caused by the protozoan parasite Leishmania donovani and transmitted to humans by the bite of infected female phlebotomine sandflies.

Q.32 — Official Correct Answer: (A) Calcium chloride ($CaCl_2$)

Question: Which of the following chemical compounds exhibits electrovalent (ionic) bonding rather than covalent bonding?
Solution Rationale: Calcium chloride ($CaCl_2$) is an ionic compound formed between a metal (calcium) and non-metal (chlorine) by complete transfer of electrons. CH_4, CCl_4, and NH_3 are covalent compounds.

Q.33 — Official Correct Answer: (A) International Court of Justice (ICJ) — The Hague (Netherlands)

Question: Which of the following pairs of international judicial or financial organizations and their headquarters is correctly matched?
Solution Rationale: The ICJ is headquartered at the Peace Palace in The Hague, Netherlands. IMF and World Bank are in Washington, D.C.; WTO is in Geneva.

Q.34 — Official Correct Answer: (B) Glow of a firefly

Question: Which one among the following natural or artificial processes produces light through a chemical reaction (chemiluminescence)?
Solution Rationale: Fireflies produce light through bioluminescence, a form of chemiluminescence where the pigment luciferin is oxidized in the presence of the enzyme luciferase and ATP without producing heat.

Q.35 — Official Correct Answer: (A) Alcohol (Ethanol)

Question: Which one among the following substances is lipid-soluble and absorbed directly and rapidly across the epithelial lining of the human stomach into the bloodstream?
Solution Rationale: Unlike most foods which require enzymatic digestion and are absorbed in the small intestine, ethanol is lipid-soluble and can be directly absorbed through the gastric mucosa of the stomach.

Q.36 — Official Correct Answer: (A) SAGA-220

Question: Which one among the following was developed as an indigenous high-performance supercomputer by the Indian Space Research Organisation (ISRO) in 2011?
Solution Rationale: SAGA-220 (Supercomputer for Aerospace with GPU Architecture-220 teraflops) was built by ISRO at the Vikram Sarabhai Space Centre (VSSC) using commercial off-the-shelf GPU hardware.

Q.37 — Official Correct Answer: (B) Sodium

Question: Which one of the following metals is stored in kerosene oil to prevent it from reacting with atmospheric moisture and oxygen?
Solution Rationale: Sodium and potassium are highly reactive alkali metals that react violently and exothermically with water and oxygen at room temperature, releasing combustible hydrogen. They are safely preserved immersed in kerosene.

Q.38 — Official Correct Answer: (B) Carbon dioxide

Question: Which one of the following gases is responsible for the effervescence observed when baking soda is added to vinegar?
Solution Rationale: When sodium bicarbonate (baking soda, $NaHCO_3$) reacts with acetic acid (vinegar, CH_3COOH), sodium acetate, water, and carbon dioxide (CO_2) gas are produced. The rapid release of CO_2 produces vigorous effervescence.

Q.39 — Official Correct Answer: (B) Iron

Question: Which one of the following elements is essential for the formation of chlorophyll in green plants?
Solution Rationale: Official UPSC CDS Solution: Assessed under General Science (Physics, Chemistry, Biology) according to UPSC CDS standards.

Q.40 — Official Correct Answer: (B) Leh

Question: Which one of the following places of India experiences highest atmospheric pressure during winter?
Solution Rationale: Official UPSC CDS Solution: Assessed under General Science (Physics, Chemistry, Biology) according to UPSC CDS standards.

Q.41 — Official Correct Answer: (C) Black hole

Question: Which one of the following is an object with such a strong gravitational field that even light cannot escape from its surface?
Solution Rationale: Official UPSC CDS Rationale: Tested in CDS General Knowledge (General Science (Physics, Chemistry, Biology)).

Q.42 — Official Correct Answer: (A) Malaria

Question: Which one of the following human diseases is caused by a protozoan parasite and NOT by bacteria or viruses?
Solution Rationale: Malaria is caused by the protozoan Plasmodium (transmitted by female Anopheles mosquito). Tetanus is bacterial; Measles and Mumps are viral diseases.

Q.43 — Official Correct Answer: (A) Tejas

Question: Which one of the following is India's indigenously designed and built single-engine supersonic Light Combat Aircraft (LCA)?
Solution Rationale: LCA Tejas is an indigenous single-engine, delta-wing multirole light fighter aircraft designed by the Aeronautical Development Agency (ADA) and manufactured by Hindustan Aeronautics Limited (HAL).

Q.44 — Official Correct Answer: (C) Magnesium

Question: Which one of the following metals cannot be used as a sacrificial anode ?
Solution Rationale: Official UPSC CDS Rationale: Tested in CDS General Knowledge (General Science (Physics, Chemistry, Biology)).

Q.45 — Official Correct Answer: (A) Potassium alum (Potash alum)

Question: Which one of the following chemical compounds is widely used as a mordant in the textile dyeing and leather tanning industries?
Solution Rationale: Potash alum (potassium aluminum sulfate) acts as a mordant by providing aluminum ions that bind natural dyes to textile fibers, fixing colors permanently.

Q.46 — Official Correct Answer: (A) National Defence Academy (NDA) — Khadakwasla (Pune)

Question: Which one of the following pairs of Indian military training academies and their locations is correctly matched?
Solution Rationale: NDA is in Khadakwasla (Pune); IMA is in Dehradun; OTA is in Chennai/Gaya; INA is in Ezhimala (Kerala).

Q.47 — Official Correct Answer: (C) Iodine

Question: Which one of the following deficiencies is most frequently responsible for anemia in India ?
Solution Rationale: Official UPSC CDS Rationale: Tested in CDS General Knowledge (Defence, Military & Current Affairs).

Q.48 — Official Correct Answer: (A) Dengue fever

Question: Which one of the following viral diseases is transmitted to humans by the bite of the female Aedes aegypti mosquito?
Solution Rationale: Aedes aegypti mosquitoes transmit Dengue, Chikungunya, Zika, and Yellow fever. Malaria is spread by Anopheles, Kala-azar by sandflies, and sleeping sickness by tsetse flies.

Q.49 — Official Correct Answer: (A) Nichrome

Question: Which one of the following alloys is commonly used for making heating elements in electrical appliances like heaters and electric irons?
Solution Rationale: Nichrome (an alloy of 80% nickel and 20% chromium) has high electrical resistivity and a high melting point, and does not easily oxidize at high temperatures.

Q.50 — Official Correct Answer: (A) 2,4-D (2,4-Dichlorophenoxyacetic acid)

Question: Which one of the following synthetic chemical compounds is widely used by farmers as a selective systemic herbicide to destroy broadleaf weeds in cereal crops?
Solution Rationale: 2,4-D is a synthetic auxin herbicide that selectively eliminates dicotyledonous (broadleaf) weeds without harming monocotyledonous grasses (wheat, corn, rice).

Q.51 — Official Correct Answer: (C) The athlete's kinetic energy

Question: Which one of the following parameters will remain constant during the fall ?
Solution Rationale: Official UPSC CDS Rationale: Tested in CDS General Knowledge (Defence, Military & Current Affairs).

Q.52 — Official Correct Answer: (A) Atomic radius

Question: Which one of the following physical properties decreases systematically as one moves from left to right across a period in the Periodic Table?
Solution Rationale: Atomic radius decreases across a period from left to right due to increased effective nuclear charge pulling valence electrons closer to the nucleus.

Q.53 — Official Correct Answer: (A) 1, 2, 3 and 4

Question: Which of the following statements about the commonly used automobile battery are true? 1. It is usually a lead-acid battery. 2. It has six cells with a potential of 2 Volt each. 3. Its cells work as Galvanic Cells while discharging power. 4. Its cells work as electrolytic cells while recharging. Select the correct answer using the code given below :
Solution Rationale: A standard 12V car battery consists of six 2V lead-acid cells connected in series. When delivering current it operates as a galvanic cell (chemical to electrical); when being recharged by the alternator it operates as an electrolytic cell.

Q.54 — Official Correct Answer: (C) Alkaloid (Caffeine / Theine)

Question: Which one of the following is responsible for the stimulating effect of tea?
Solution Rationale: The stimulating effect of tea is primarily due to the purine alkaloid caffeine (also known as theine), which acts as a central nervous system stimulant.

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